STAT 425 · Fall 2026

Bayesian Inference — Final Course Project

Purpose

In this project, you will investigate a question of interest to science, industry, the public, or yourself using Bayesian inference. Use the tools learned in this course; methods beyond the course are welcome but not required. The goal is to practice the full process of applying statistical ideas to a meaningful question, from planning to communicating findings.

Teams and timeline

Work individually or form your own team of 2 or 3 students. Submit the names and NetIDs of all team members by Friday, November 6. Your one-page proposal should also list all team members. Each team submits one proposal, one presentation, and one final report. All members should contribute meaningfully.

Fri, Nov 6	**Team roster due: submit names and NetIDs only; no topic or proposal is required.**
Mon, Nov 16	One-page proposal due (team roster, question, dataset, and initial plan). Early submission welcome.
Mon, Nov 30 & Wed, Dec 2	Team presentations; approximately 5–8 minutes per team. Exact time depends on the number of teams and will be announced in advance.
Fri, Dec 4	Final report, code, and AI-use record due.

Thanksgiving recess (Nov 25–27): no class meetings or project deadlines. Presentation dates may be adjusted to the number of teams. Check Canvas for submission times.

Proposal (no more than one page)

Give your project title; team member names and NetIDs; the question(s) you hope to answer and why they matter; the dataset, source, and key variables; and a tentative Bayesian approach (model, prior assumptions, and intended analysis). Briefly explain the plan for completing the project. A model need not be fully developed at the proposal stage. You may revise your topic as the project develops.

Final report (no more than 10 pages)

Submit a PDF or DOCX report. References, code appendix, AI-use record, and individual self-reflections do not count toward the page limit. Include the following components:

1. Statement of the problem. State the purpose of your project, the question(s) you aim to answer, and effects or quantities of interest.

2. Introduction and background. Summarize relevant background, your motivation, and the approach you planned.

3. Data collection and description. Describe data sources, variables, sample size, and appropriate descriptive summaries or visualizations.

4. Data analysis and results. Develop and examine Bayesian models that address your questions; describe the fitted model and reasoning, posterior results and their uncertainty, model fit, and (where feasible) sensitivity to priors or assumptions. Use well-organized tables and plots rather than pages of raw software output.

5. Summary and discussion. Summarize findings, limitations, and possible future work.

6. Appendix / executable code. Supply clearly commented R or Python code and the necessary data or instructions for reproducibility.

7. References and acknowledgments. Cite datasets, articles, software, and other sources and explain material external assistance.

8. Individual self-reflection. Each member should briefly describe time spent, contributions, what was learned, challenges, what they would do differently, and advice for future students.

9. AI use and prompt record. See the policy below.

AI use and transparency

AI tools may be used for learning, brainstorming, background research, coding help, debugging, visualization, and writing support. Verify their outputs; you are responsible for your project's originality, statistical reasoning, calculations, and citations. In the final submission, disclose which tools you used, for what purposes, and how you checked their output. Include a record of the substantive prompts used for the project (or a shareable conversation transcript/link where permitted), organized in an appendix; routine spelling or grammar prompts need not be listed individually. Do not upload confidential, personal, or restricted data to public AI tools. Cite original sources rather than relying on AI-generated citations.

Presentation

Each team will have approximately 5–8 minutes to present. The exact time will depend on the number of teams and will be announced in advance. Focus on a clear question, model, central findings and uncertainty, and what you learned. Readable plots and concise interpretation are preferable to extensive software output. All team members should participate.

Alternative project formats

You may study a Bayesian or computational method through theory, simulation, or visualization rather than analyzing an external dataset. Examples include prior sensitivity, robustness to model misspecification, properties of Bayesian estimators, or an interactive app. Explain the substantive Bayesian question and how your work evaluates it.

Evaluation

Projects will be evaluated using the approach from prior courses: meeting the stated timeline; interest and clarity of the problem; clarity and completeness of the report; technical correctness and quality of the analysis; presentation; and notable features such as thoughtful data collection, visualization, or an interactive tool. The project counts for 30% of the course grade.

Possible topics and past project examples

Possible areas and data sources:

- Rice- or Houston-related problems and datasets.
- Data collected by you, including surveys, social media, or other publicly available sources (subject to privacy and ethical considerations).
- Public data platforms such as Kaggle.
- Sports statistics.
- U.S. statistics and public datasets on social, demographic, or economic topics.
- Financial statistics, such as publicly available market data.
- Housing and real-estate data; for example, the Ames housing dataset discussed in De Cock (2011).

Examples of past student project topics:

1. What Houston Floods Tell Us: An Analysis of Memorial Day Flood and Tax Day Flood
2. Determining the Relationship between Psychological Distress and Drug Abuse/Sociodemographic Factors
3. Course Ideals among Rice Students
4. Predicting Voter Turnout in Rice University Presidential Elections to Improve Campaign Directed Voter Targeting
5. Weather Elements that Influence San Francisco Bay Area Bike Usage
6. Why Do More and More People Choose to Live Alone?
7. An Investigation of the Perceptions of Password Security
8. Are We Able to Predict How Many Opioid Prescriptions a Doctor Will Give Annually?
9. Do Liberal Arts Students Earn More Than STEM Students in the Long Run?
10. (Statistically) Significant Others: The Science of Speed Dating
11. Multivariate Linear Regression Model on Departure Delay Time
12. Did Harvey Hit on Stock Price: What to Look?
13. Relationship between Nutrition and Food and GPA in Students at Mercyhurst University
14. From Film to Data: Models for Predicting IMDb Scores and Gross Revenue

15. What Determines a Song’s Popularity?
16. College Rankings: Understanding Determining Factors
17. Complications in the Treatment and Healing of Severe Burns
18. Predicting Board Game Ratings
19. Analysis of Meme Trends on Twitter
20. Factors in Evaluation of Course Quality Among Rice Students
21. Understanding Houston Zillow Listings via Census Data